


# 3

NUMBER AND ALGEBRA

# WHOLE NUMBERS

We use numbers all the time: for measuring fabric to make a quilt or an item of clothing; measuring the temperature, rainfall or time; budgeting how we will spend our money; setting up training at the gym or keeping score in all different sports.




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## Chapter outline

| | Working mathematically | | | | |
|---|---|---|---|---|---|
| **3.01** Rounding and estimating | U | F | PS | R | C |
| **3.02** Multiplying numbers | U | F | PS | R | |
| **3.03** Dividing numbers | U | F | PS | R | |
| **3.04** Divisibility tests | U | F | | R | C |
| **3.05** Long division* | U | F | PS | | |
| **3.06** Roman numerals* | U | F | | | C |
| **3.07** Powers and index notation | U | F | | R | C |
| **3.08** Square root and cube root | U | F | | R | C |
| **3.09** Prime and composite numbers | U | F | | R | C |
| **3.10** Factor trees | U | F | | R | |
| **3.11** Highest common factor | U | F | | R | C |
| **3.12** Lowest common multiple* | U | F | | R | C |

*EXTENSION

## Wordbank

**composite number** A number with more than 2 factors

**divisibility test** A rule for testing whether a number is divisible by a specific alue, for example, divisible by 3

**factor** A value that divides evenly into a given number, for example, 3 is a factor of 15

**factor tree** A diagram that lists the prime factors of a number

**highest common factor (HCF)** The largest factor shared by 2 or more numbers, also called the **greatest common divisor (GCD)**

**index notation** A way of writing powers for the repeated multiplication of a number, for example, $3^5$

**prime number** A number with only 2 factors, 1 and itself

**square root (of a number, symbol $\sqrt{}$)** The positive value which, if squared, gives the number

**Quiz**
Wordbank 3

**U** = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

## Videos (10):

**3.01** Estimating answers
**3.02** Long multiplication • Mental multiplication
**3.03** Dividing numbers • Mental division
**3.04** Divisibility tests
**3.10** Factor trees • Using factor trees to find square roots
**3.11** Highest common factor • HCF and factor trees

## *Twig* videos (5):

**3.01** Rounding: snails vs rockets
**3.06** The Romans and numerals • Numbers: Life without numbers
**3.07** The biggest number ever
**3.09** The prime number code

## *PhET* interactives (3):

**3.02** Arithmetic • Area model Introduction • Area model Multiplication

## Quizzes (6):

- Wordbank 3
- SkillCheck 3
- Mental skills 3A
- Mental skills 3B
- Language of maths 3
- Test yourself 3

## Skillsheets (8):

**3.01** Place value • Reading and writing large numbers • Rounding whole numbers
**3.04** Factors and divisibility
**3.06** Roman numerals
**3.08** Square roots and cube roots
**3.09** Prime and composite numbers
**3.10** Prime factors by repeated division

## Worksheets (13):

**SkillCheck** Calculation aids • Number grids
**3.02** Whole numbers 6 • Calculation aids • Number grids
**3.03** Whole numbers 7
**3.04** Divisibility tests
**3.05** Four operations
**3.07** Big numbers
**3.08** Powers and roots
**3.09** Sieve of Eratosthenes
**3.10** Factor trees
**3.12** Perfect and amicable numbers
**Mind map** Whole numbers

## Puzzles (7):

**SkillCheck** Cross number puzzle • Magic squares
**3.02** Cross number puzzle • Magic squares • Find the quote 2
**3.03** The accidental detective
**3.06** Roman numerals dominoes
**3.08** Square root Snap
**Language of maths** Number find-a-word

## Presentation (1):

**3.11** Highest common factors

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

## In this chapter you will:

- round whole numbers, estimate answers and check whether answers sound reasonable
- solve problems involving multiplying and dividing whole numbers
- understand and use divisibility tests
- (EXTENSION) use long division to divide by a 2-digit number
- (EXTENSION) interpret Roman numerals
- recognise index notation, powers, square roots and cube roots
- recognise prime and composite numbers
- use a factor tree to write a whole number as a product of its prime factors
- find highest common factors (HCF) of 2 or more numbers
- (EXTENSION) find lowest common multiples (LCM) of 2 or more numbers

## SkillCheck ANSWERS ON P. 567

**1** Is 691 odd or even? How can you tell?

**2** Is 270 divisible by 5? How can you tell?

**3** Evaluate each product.

| | | | |
|---|---|---|---|
| **a** $7 \times 3$ | **b** $6 \times 5$ | **c** $4 \times 8$ | **d** $3 \times 6$ |
| **e** $9 \times 4$ | **f** $5 \times 8$ | **g** $7 \times 7$ | **h** $6 \times 9$ |
| **i** $8 \times 8$ | **j** $9 \times 7$ | **k** $4 \times 6$ | **l** $7 \times 8$ |
| **m** $-3 \times (-3)$ | **n** $9 \times 9$ | **o** $5 \times 5$ | **p** $4 \times (-4)$ |

**4** Write all the factors of:

**a** 20 **b** 12 **c** 19

**5** Is 166 divisible by 2? How can you tell?

**6** Evaluate each quotient.

| | | | |
|---|---|---|---|
| **a** $24 \div 3$ | **b** $35 \div 7$ | **c** $24 \div 4$ | **d** $36 \div 6$ |
| **e** $20 \div 5$ | **f** $72 \div 9$ | **g** $48 \div 8$ | **h** $32 \div 4$ |
| **i** $28 \div 7$ | **j** $45 \div 5$ | **k** $54 \div 9$ | **l** $63 \div 7$ |

**7** Write the first 5 multiples of:

**a** 8 **b** 7 **c** 12

**8** Is 34 divisible by 10? How can you tell?

**Quiz**
SkillCheck 3

**Worksheets**
Calculation aids
Number grids

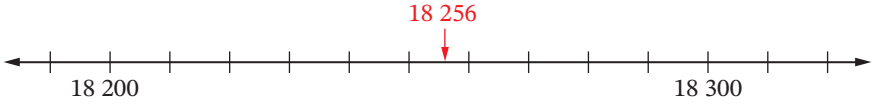
**Puzzles**
Cross number puzzle
Magic squares



# Rounding and estimating 3.01

## Rounding numbers

When calculating with numbers, it is often useful to have a rough idea of the answer before actually working it out. Rounded numbers are easier to work with and to compare. For example, a town's population of 18 256 can be rounded to 18 300 (to the nearest hundred) because 18 256 is closer to 18 300 than 18 200.




**Skillsheets**
Place value
Reading and writing large numbers
Rounding whole numbers

Video
Rounding Snails vs rockets

## Rounding whole numbers

To round a number, 'cut' it at the required place and look at the **digit** in the next place:

- if the digit is less than 5 (that is 0, 1, 2, 3 or 4), **round down**
- if the digit is 5 or more (that is 5, 6, 7, 8 or 9), **round up**

### Example 1

Round 8470 to the nearest hundred.

#### Solution

Counting by hundreds, 8470 is between 8400 and 8500.

- In 8**4**70, the hundreds digit is **4**
- The digit in the next (tens) place, 7, is more than 5, so **round up** to 8500.

$8470 \approx 8500$ (rounded to the nearest hundred). The symbol '≈' means 'approximately equal to'.

### Example 2

Round 247 182 to the nearest thousand.

#### Solution

In 247 182, the thousands digit is 7 and the next digit is 1.

$1 < 5$, so round down to 247 000. The thousands digit, 7, stays the same.

$247\,182 \approx 247\,000$ (rounded to the nearest thousand).

## Estimating answers

A quick way of estimating an answer is to round each number in the calculation.

Video
Estimating answers

### Example 3

Estimate the answer to each expression.

**a** $631 + 280 + 51 + 43 + 96$ **b** $67 \times 12$

**c** $55 + 132 - 34 + 17 - 78$ **d** $510 \div 24$

#### Solution

**a** $631 + 280 + 51 + 43 + 96 \approx 600 + 300 + 50 + 40 + 100$ Estimating

$\approx (600 + 300 + 100) + (50 + 40)$

$= 1000 + 90$

$= 1090$ (Exact answer = 1101)

**b** $67 \times 12 \approx 70 \times 10$ Estimating

$= 700$ (Exact answer = 804)

**c** $55 + 132 - 34 + 17 - 78 \approx 60 + 130 - 30 + 20 - 80$ Estimating
$= (60 + 20 - 80) + (130 - 30)$
$= 0 + 100$
$= 100$ (Exact answer = 92)

**d** $510 \div 24 \approx 500 \div 20$ Estimating
$= 50 \div 2$
$= 25$ (Exact answer = 21.25)



## EXERCISE 3.01 ANSWERS ON P. 568

### Rounding and estimating U F PS R C

EXAMPLE 1

**1** Round each number to the nearest hundred. C

**a** 3148 **b** 49 028 **c** 2597 **d** 4 934 277

EXAMPLE 2

**2** Round each number to the nearest thousand. C

**a** 23 538 **b** 45 370 **c** 62 941 **d** 47 929

**3** Round each number to the nearest ten. C

**a** 45 819 **b** 1699 **c** 8314 **d** 71 262

**4** Round 64 218 to the nearest: C

**a** thousand **b** hundred **c** ten **d** ten thousand

**5** Round 1 327 509 to the nearest: C

**a** thousand **b** ten thousand **c** ten **d** million

EXAMPLE 3

**6** Estimate the answer for each expression. R

**a** $27 + 11 + 87 + 142 + 64$ **b** $55 + 34 - 22 - 46 + 136$
**c** $684 + 903$ **d** $35 + 81 + 110 + 22 + 7$
**e** $517 - 96$ **f** $210 - 38 - 71 + 151 - 49$
**g** $766 - 353$ **h** $367 \times 2$
**i** $83 \times 81$ **j** $984 \times 16$
**k** $828 \div 3$ **l** $507 \div 7$

**7** Over the holidays, 27 792 people visited a museum. Write this figure correct to the nearest hundred. C

**8** The extensions on Nina's house are quoted as costing $17 464. Write this amount correct to the nearest $100. C

**9** The crowd at the football grand final was 104 427. Round this figure to the nearest thousand. C

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**10** Max buys bags of washers to use in his plumbing business. Each bag contains between 48 and 51 washers. Max has 11 bags in his supply cupboard. Approximately how many washers does he have? Select **A**, **B**, **C** or **D**. PS R

**A** 50 **B** 550 **C** 1100 **D** 2500

**11** The distance between Brisbane and Sydney is 998 km. Round this distance to the nearest: C

**a** 10 km **b** 100 km **c** 1000 km

**12** The population of Australia is 26 921 840. Round this figure to the nearest thousand. C

**13** Write a number that can be rounded to: R C

**a** 370 **b** 5400 **c** 12 900 **d** 6000

# 3.02 Multiplying numbers

**Video**
Long multiplication

**Worksheets**
Whole numbers 6
Calculation aids
Number grids

**Puzzles**
Cross number puzzle
Magic squares
Find the quote 2

**Interactives**
Arithmetic
Area model Introduction
Area model Multiplication

## Example 4

Evaluate each product.

**a** $243 \times 6$ **b** $573 \times 36$

### Solution

**a**
$$\begin{array}{r} 243 \times \\ 6 \\ \hline 1458 \end{array}$$

$243 \times 6 \approx 200 \times 6$ Check by estimating
$= 1200$

**b**
$$\begin{array}{r} 573 \times \\ 36 \\ \hline 3438 \\ 17\,190 \\ \hline 20\,628 \end{array}$$

$573 \times 36 \approx 600 \times 40$ Check by estimating
$= 24\,000$

## Mental multiplication

| Multiplying by: | Mental strategy |
|---|---|
| 2, 4 or 8 | Double once, 2 times or 3 times respectively |
| 5 | Halve, then multiply by 10 (because $\frac{1}{2} \times 10 = 5$) |
| 9 | Multiply by 10, then subtract the number |
| 10 | Insert 0 at the end of the number |
| 100 | Insert 00 at the end of the number |

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## Example 5

Evaluate each product mentally.

**a** $68 \times 4$ **b** $36 \times 5$ **c** $12 \times 9$ **d** $14 \times 8$

### Solution

**a**
$68 \times 4 = 68 \times 2 \times 2$ Double twice
$68 \times 2 = (60 + 8) \times 2$ Double 68
$= 60 \times 2 + 8 \times 2$
$= 120 + 16$
$= 136$
$136 \times 2 = (130 + 6) \times 2$ Double 136
$= 130 \times 2 + 6 \times 2$
$= 260 + 12$
$= 272$
$\therefore 68 \times 4 = 272$ Estimate: $68 \times 4 \approx 70 \times 4 = 280$

**b**
$36 \times 5 = 36 \times \frac{1}{2} \times 10$ Because $\frac{1}{2} \times 10 = 5$
$= 18 \times 10$ Insert a 0 at the end
$= 180$ Estimate: $36 \times 5 \approx 40 \times 5 = 200$

**c**
$12 \times 9 = 12 \times 10 - 12$ Multiply by 10, then subtract the number
$= 120 - 12$ Estimate: $12 \times 9 \approx 12 \times 10 = 120$
$= 108$

**d**
$14 \times 8 = 14 \times 2 \times 2 \times 2$ Double 3 times
$14 \times 2 = (10 + 4) \times 2$ Double 14
$= 10 \times 2 + 4 \times 2$
$= 20 + 8$
$= 28$
$28 \times 2 = (20 + 8) \times 2$ Double 28
$= 20 \times 2 + 8 \times 2$
$= 40 + 16$
$= 56$
$56 \times 2 = (50 + 6) \times 2$ Double 56
$= 50 \times 2 + 6 \times 2$
$= 100 + 12$
$= 112$
$\therefore 14 \times 8 = 112$ Estimate: $14 \times 8 \approx 14 \times 10 = 140$



**Video**
Mental multiplication

3.02

## EXERCISE 3.02 ANSWERS ON P. 568

## Multiplying numbers U F PS R

EXAMPLE 4

**1** Copy and complete each multiplication.

**a** $34 \times 7$ **b** $219 \times 5$ **c** $28 \times 16$ **d** $325 \times 21$

**2** Jesinta pays $15 each time she plays tennis. How much does Jesinta pay to play 42 times a year?

**3** Evaluate each product.

**a** $14 \times 8$ **b** $238 \times 3$ **c** $344 \times 9$ **d** $506 \times 7$

**e** $1084 \times 3$ **f** $45 \times 14$ **g** $64 \times 25$ **h** $107 \times 32$

**4** A bus route is 46 km long. A bus makes 12 trips in one day. Find the total distance travelled each day. PS

**5** Ned can type 76 words per minute. How many words can he type in 15 minutes? PS

EXAMPLE 5

**6** Estimate and then evaluate each product mentally. R

**a** $85 \times 2$ **b** $39 \times 2$ **c** $64 \times 2$ **d** $57 \times 4$

**e** $28 \times 4$ **f** $44 \times 4$ **g** $16 \times 8$ **h** $33 \times 8$

**7** Estimate and then evaluate each product mentally. R

**a** $22 \times 5$ **b** $14 \times 5$ **c** $28 \times 5$ **d** $36 \times 5$

**e** $54 \times 5$ **f** $82 \times 5$ **g** $16 \times 5$ **h** $48 \times 5$

**8** Estimate and then evaluate each product mentally. R

**a** $34 \times 9$ **b** $51 \times 9$ **c** $27 \times 9$ **d** $19 \times 9$

**e** $45 \times 10$ **f** $18 \times 100$ **g** $36 \times 10$ **h** $41 \times 100$

**9** How many hours are there in one week? Select the correct answer **A**, **B**, **C** or **D**. PS R

**A** 84 **B** 151 **C** 168 **D** 240

**10** Mala ordered 4 pallets of oranges to sell. Each pallet holds 8 boxes of oranges. Each box contains 124 oranges. PS R

**a** How many boxes of oranges did Mala order?

**b** How many oranges is this?

**c** The oranges are sold for $52 per box. How much money will Mala receive if she sells all boxes?

# Dividing numbers

3.03

Division is the opposite of multiplication.

## Example 6

Evaluate each quotient.

**a** $318 \div 6$ **b** $1964 \div 5$

### Solution

**a**
$$6\overline{)3\,1^{1}\,8}\quad 53$$

$318 \div 6 = 53$

**b**
$$5\overline{)19^{4}6^{1}4}\quad 3\;\;9\;\;2\text{ r}4$$

Write the remainder, 4, as a fraction over the divisor, 5:

$1964 \div 5 = 392\frac{4}{5}$





**Worksheet**
Whole numbers 7

**Puzzle**
The accidental detective



**Video**
Dividing numbers

## Mental division

| Dividing by: | Mental strategy |
|---|---|
| 2, 4 or 8 | Halve once, 2 times or 3 times respectively |
| **Dividing a multiple of 10 by:** | |
| 5 | Divide by 10, then double |
| 10 | Delete 0 from the end of the number |
| 100 | Delete 00 from the end of the number |

## Example 7

Evaluate each quotient mentally.

**a** $520 \div 8$ **b** $400 \div 100$ **c** $260 \div 5$ **d** $316 \div 4$

### Solution

**a** $520 \div 8 = 520 \div 2 \div 2 \div 2$ Halve 3 times

$520 \div 2 = (500 + 20) \div 2$ Halve 520

$= 500 \div 2 + 20 \div 2$

$= 250 + 10$

$= 260$



**Video**
Mental division

$260 \div 2 = (200 + 60) \div 2$ — Halve 260

$= 200 \div 2 + 60 \div 2$

$= 100 + 30$

$= 130$

$130 \div 2 = (100 + 30) \div 2$ — Halve 130

$= 100 \div 2 + 30 \div 2$

$= 50 + 15$

$= 65$

$\therefore 520 \div 8 = 65$ — Estimate:

$520 \div 8 \approx 520 \div 10 = 52$

**b** $400 \div 100 = 4 \div 1 = 4$ — Delete 00 from the end of the number

**c** $260 \div 5 = 260 \div 10 \times 2$ — Divide by 10, then double

$= 26 \times 2$ — Estimate:

$= 52$ $260 \div 5 \approx 300 \div 5 = 60$

**d** $316 \div 4 = 316 \div 2 \div 2$ — Halve 2 times

$316 \div 2 = (300 + 16) \div 2$ — Halve 316

$= 300 \div 2 + 16 \div 2$

$= 150 + 8$

$= 158$

$158 \div 2 = (150 + 8) \div 2$ — Halve 158

$= 150 \div 2 + 8 \div 2$

$= 75 + 4$

$= 79$

$\therefore 316 \div 4 = 79$ — Estimate:

$316 \div 4 \approx 320 \div 4 = 80$

## EXERCISE 3.03 ANSWERS ON P. 568

### Dividing numbers U F PS R

EXAMPLE 6

**1** Evaluate each quotient.

**a** $44 \div 4$ **b** $68 \div 2$ **c** $84 \div 7$ **d** $210 \div 3$

**e** $105 \div 7$ **f** $390 \div 5$ **g** $441 \div 3$ **h** $861 \div 7$

**2** Evaluate each quotient, expressing the remainder as a fraction.

**a** $49 \div 4$ **b** $688 \div 3$ **c** $952 \div 6$ **d** $1815 \div 7$

**3** At Westvale Catholic College, there are 135 students in Year 7. If they are placed evenly into 5 classes, how many students are in each class?
Select the correct answer **A**, **B**, **C** or **D**. PS R

**A** 21 **B** 25 **C** 27 **D** 29

**4** Divide a restaurant bill of $342 evenly among 6 people.

**5** Evaluate each quotient mentally. R

**a** 320 ÷ 10 **b** 8000 ÷ 100 **c** 60 000 ÷ 100 **d** 1800 ÷ 10
**e** 410 ÷ 5 **f** 90 ÷ 5 **g** 230 ÷ 5 **h** 600 ÷ 5

**6** Evaluate each quotient mentally. R

**a** 144 ÷ 2 **b** 232 ÷ 2 **c** 648 ÷ 2 **d** 970 ÷ 2
**e** 216 ÷ 4 **f** 488 ÷ 4 **g** 872 ÷ 8 **h** 208 ÷ 8

**7** A factory makes 1232 boxes of cereal. These are packed into cartons that each hold 8 boxes of cereal. How many cartons are needed to pack these boxes? PS R

EXAMPLE 7

3.04

# Divisibility tests

3.04

Is 840 divisible by 3? How do you know?

It is often useful to know whether a number is divisible by another number or not. Numbers that divide evenly (without a remainder) into a bigger number are called the **factors** or **divisors** of the number. **Divisibility tests** are rules for finding whether a number is divisible by specific values.

**Worksheet** Divisibility tests

**Skillsheet** Factors and divisibility

## Divisibility tests

| Divisible by: | Test |
|---|---|
| 2 | Last digit 0, 2, 4, 6 or 8 |
| 3 | Sum of digits divisible by 3 |
| 4 | Last 2 digits form a number divisible by 4<br>Or: Sum of double the tens digit and the units digit divisible by 4 |
| 5 | Last digit 0 or 5 |
| 6 | Divisible by 2 and 3 |
| 8 | Last 3 digits form a number divisible by 8 |
| 9 | Sum of digits divisible by 9 |
| 10 | Last digit 0 |

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Video
Divisibility tests

## Example 8

Test whether 2016 is divisible by:

**a** 3 **b** 4 **c** 6 **d** 8

### Solution

**a** Sum of digits = 2 + 0 + 1 + 6 = 9,
which is divisible by 3.
So 2016 is divisible by 3. (Check: 2016 ÷ 3 = 672)

**b** The last 2 digits of 2016 are 16.
16 is divisible by 4.
So 2016 is divisible by 4.
OR: Sum of double the tens digit and the units digit = 2 × 1 + 6 = 8.
8 is divisible by 4.
So 2016 is divisible by 4. (Check: 2016 ÷ 4 = 504)

**c** 2016 ends in 6, so it is divisible by 2 (even).
2016 is divisible by 3 (from part **a**).
So 2016 is divisible by 6 (because it is divisible by 2 and 3). (Check: 2016 ÷ 6 = 336)

**d** The last 3 digits of 2016 are 016 = 16.
16 is divisible by 8.
So 2016 is divisible by 8. (Check: 2016 ÷ 8 = 252)

*Note*:

- If a number is divisible by 6, it must also be divisible by 2 and 3 (because 2 × 3 = 6)
- If a number is divisible by 10, it must also be divisible by 2 and 5 (because 2 × 5 = 10)
- If a number is divisible by 4, it must also be divisible by 2 (because 2 × 2 = 4)
- If a number is divisible by 9, it must also be divisible by 3 (because 3 × 3 = 9)
- If a number is divisible by 8, it must also be divisible by 2 and 4 (because 2 × 4 = 8)
- If a number is divisible by another number, it must also be divisible by the **factors** of that number

**EXERCISE 3.04** ANSWERS ON P. 568

## Divisibility tests UFRC

EXAMPLE 8

**1** Test whether each number is divisible by 3.

**a** 140 **b** 612 **c** 315 **d** 928
**e** 209 **f** 525 **g** 132 **h** 1652

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9780170477963

**2** Test whether each number is divisible by 2.

**a** 117 **b** 205 **c** 6196 **d** 340

**3** Test whether each number is divisible by 6. Note that the numbers in parts **a** to **d** have already been tested for divisibility by 3 in question **1**.

**a** 140 **b** 612 **c** 315 **d** 928

**e** 475 **f** 303 **g** 864 **h** 1278

**4** Test whether each number is divisible by 4.

**a** 2040 **b** 518 **c** 365 **d** 242

**e** 356 **f** 728 **g** 4176 **h** 817

**5** Test whether each number is divisible by 9.

**a** 812 **b** 309 **c** 567 **d** 243

**e** 837 **f** 462 **g** 6444 **h** 3111

**6** Test whether 1964 is divisible by 8.

**7** Explain by just looking at the last digit why: R C

**a** 409 is not divisible by 4 **b** 316 is not divisible by 5

**c** 2015 is not divisible by 8 **d** 343 is not divisible by 6

**e** 472 is not divisible by 10 **f** 511 is not divisible by 12

**8** Which number is divisible by both 4 and 5? Select the correct answer **A**, **B**, **C** or **D**. R

**A** 10 **B** 15 **C** 20 **D** 25

**9** If a number is divisible by 3 and 4, what other number must it also be divisible by? R

**10** Write a number between 50 and 100 that is divisible by: R

**a** 3 and 5 **b** 4 and 5 **c** 6 and 7 **d** 2 and 6

**11** Which number is divisible by 6, 8 and 9? Select **A**, **B**, **C** or **D**. R

**A** 48 **B** 102 **C** 126 **D** 144

**12** There are several complicated divisibility tests for 7. Here is the most popular one:

- Remove the last digit of the number and double it
- Subtract it from the rest of the number
- If the difference is 0 or divisible by 7, the number is divisible by 7

For example, to test 112:

Last digit = 2, double 2 = 4

11 – 4 = 7, which is divisible by 7

So 112 is divisible by 7.

Test whether each number is divisible by 7. R

**a** 357 **b** 1013 **c** 6258 **d** 901

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**13** Here is a divisibility test for 11.

- Add the digits in the odd places of the number
- Add the digits in the even places of the number
- Subtract the 2 sums
- If the difference is 0 or divisible by 11, the number is divisible by 11

For example, to test 5808:

Add digits in odd places: $5 + 0 = 5$

Add digits in even places: $8 + 8 = 16$

$16 - 5 = 11$, which is divisible by 11

So 5808 is divisible by 11.

Test whether each number is divisible by 11. R

**a** 814 **b** 2728 **c** 4051 **d** 6105

Quiz
Mental skills 3A

## Mental skills 3A: Maths without calculators ANSWERS ON P. 568

### Multiplying by a multiple of 10

Place value allows us to simply add 0s to the end of a number whenever we multiply by a power of 10.

**1** Consider these examples.

**a** $37 \times 10 = 370$ **b** $45 \times 100 = 4500$

**c** $16 \times 1000 = 16\,000$ **d** $100 \times 1000 = 100\,000$

**e** $7 \times 90 = 7 \times 9 \times 10 = 63 \times 10 = 630$

**f** $5 \times 400 = 5 \times 4 \times 100 = 20 \times 100 = 2000$

**g** $12 \times 300 = 12 \times 3 \times 100 = 36 \times 100 = 3600$

**h** $40 \times 800 = 4 \times 10 \times 8 \times 100 = 4 \times 8 \times 10 \times 100 = 32 \times 100 = 32\,000$

**2** Now evaluate each product.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | $18 \times 100$ | **b** | $26 \times 1000$ | **c** | $77 \times 10\,000$ | **d** | $10 \times 100$ |
| **e** | $315 \times 1000$ | **f** | $1000 \times 1000$ | **g** | $294 \times 10$ | **h** | $475 \times 100$ |
| **i** | $3 \times 80$ | **j** | $8 \times 200$ | **k** | $6 \times 50$ | **l** | $7 \times 30$ |
| **m** | $2 \times 6000$ | **n** | $11 \times 900$ | **o** | $4 \times 400$ | **p** | $5 \times 700$ |
| **q** | $5 \times 80$ | **r** | $25 \times 20$ | **s** | $300 \times 60$ | **t** | $900 \times 4000$ |

# Extension: Long division

3.05

**Long division** is a pen-and-paper method for dividing by numbers with 2 or more digits, that is, a number greater than 10.

EXTENSION



**Worksheet**
Four operations



**Video**
Dividing numbers



## Example 9

Use long division to evaluate each quotient.

**a** $312 \div 12$ **b** $296 \div 21$

### Solution

**a**

$$\begin{array}{r} 26 \\ 12\overline{)\,312} \\ -24\downarrow \\ \hline 72 \\ -72 \\ \hline 0 \end{array}$$

12 into 31 is 2, remainder 7

12 into 72 is 6

OR

$$\begin{array}{r} 26 \\ 12\overline{)\,312} \\ -120 \quad \text{10 times} \\ \hline 192 \\ -120 \quad \text{10 times} \\ \hline 72 \\ -72 \quad \text{6 times} \\ \hline 0 \quad \text{26 times} \end{array}$$

Guessing with 'easy' multiples of 12

Check by estimating:

$312 \div 12 = 26.$

$312 \div 12 \approx 300 \div 10 = 30$

**b**

$$\begin{array}{r} 14 \\ 21\overline{)\,296} \\ -21\downarrow \\ \hline 86 \\ -84 \\ \hline 2 \end{array}$$

21 into 29 is 1, remainder 8

21 into 86 is 4, remainder 2

OR

$$\begin{array}{r} 14 \\ 21\overline{)\,296} \\ -210 \quad \text{10 times} \\ \hline 86 \\ -84 \quad \text{4 times} \\ \hline 2 \quad \text{14 times} \end{array}$$

EXTENSION

$296 \div 21 = 14\frac{2}{21}$ Write the remainder as the numerator of a fraction

Check by estimating:

$296 \div 21 \approx 300 \div 20 = 15$

(This means $296 = 14 \times 21 + 2$) 2 is the remainder.

## EXERCISE 3.05 ANSWERS ON P. 568

### Long division U F PS

EXAMPLE 9

**1** Use long division to evaluate each quotient.

**a** $180 \div 15$ **b** $462 \div 22$ **c** $731 \div 17$

**d** $275 \div 11$ **e** $2352 \div 16$ **f** $78 \div 13$

**g** $900 \div 25$ **h** $667 \div 23$ **i** $6848 \div 32$

**2** Evaluate each quotient, writing the remainder as a fraction.

**a** $304 \div 12$ **b** $505 \div 14$ **c** $990 \div 26$

**3** At a party 275 lollies are shared equally among 25 children. How many lollies does each child get? PS

**4** A piece of wood 390 cm in length is to be cut into 15 equal pieces. How long is each piece? PS

**5** Mrs Kaur needs $2300 to purchase a new home cinema system. If she can save $92 each week, how long will it take her to save enough money to purchase the system? PS

# 3.06 Extension: Roman numerals

EXTENSION

Skillsheet Roman numerals

Puzzle Roman numerals dominoes

Video The Romans and numerals

In 117 CE, the Roman Empire ruled all of Europe, north Africa and the Middle East, and Roman numerals were widely used in western civilisation until the end of the 16th century, when they were gradually replaced by our modern Hindu-Arabic numerals. Roman numerals use letters of the alphabet and are still often seen today.

The ancient Romans used the following numerals:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| I | II | III | IV | V | VI | VII | VIII | IX | X |

| 20 | 30 | 40 | 50 | 60 | 90 | 100 | 400 | 500 | 1000 |
|---|---|---|---|---|---|---|---|---|---|
| XX | XXX | XL | L | LX | XC | C | CD | D | M |

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The Romans had an unusual method of writing numbers involving 4 or 9, to avoid writing the same letter more than 3 times in a row. A smaller number written before a larger number meant a subtraction ('minus').

- Instead of writing 4 as IIII, they wrote **IV** meaning V – I (that is 5 – 1 = 4).
- Instead of writing 9 as VIIII they wrote **IX** meaning X – I (that is 10 – 1 = 9).
- For 40, they wrote **XL** (that is 50 – 10 = 40).
- For 90, they wrote **XC** (that is 100 – 10 = 90).



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## Example 10

Write each number in Roman numerals.

**a** 23 **b** 46 **c** 279

### Solution

**a** 23 = 20 + 3
= XX + III
= XXIII

**b** 46 = 40 + 6
= XL + VI
= XLVI

**c** 279 = 200 + 70 + 9
= CC + LXX + IX
= CCLXXIX

## Example 11

Write each Roman numeral as a number.

**a** LXXIII **b** CI **c** CDXXVIII

### Solution

**a** LXXIII = LXX + III
= 70 + 3
= 73

**b** CI = C + I
= 100 + 1
= 101

**c** CDXXVIII = CD + XX + VIII
= 400 + 20 + 8
= 428

**EXERCISE 3.06** ANSWERS ON P. 568

## Roman numerals U F C

**1** Write each number using Roman numerals. C

EXAMPLE 10

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | 27 | **b** | 52 | **c** | 79 | **d** | 93 |
| **e** | 2600 | **f** | 314 | **g** | 928 | **h** | 3476 |
| **i** | 165 | **j** | 230 | **k** | 2482 | **l** | 109 |

**2** Titus, a student in ancient Rome, wrote these numerals. Rewrite them as modern numerals. C

EXAMPLE 11

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | XXVI | **b** | XL | **c** | CCLXIV | **d** | LIV |
| **e** | LXXVII | **f** | XLV | **g** | CCCXXXIX | **h** | DXXVIII |
| **i** | MMCLXII | **j** | MCMXC | **k** | XCVIII | **l** | MDVII |

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EXTENSION

**3 a** Where do you see Roman numerals being used today?

**b** Why do you think Roman numerals are no longer widely used? C

**4** The Roman word for hundred was 'centum', which is why C stands for 100. List some words beginning with 'cent' that mean one hundred of something. C

**5 a** Write this year in Roman numerals.

**b** Which year of the 21st century will use the most letters when written in Roman numerals? C

**6** Find out what the Roman numerals $\overline{V}$ and $\overline{X}$ mean. C

Video
Numbers: Life without numbers

## Investigation

### Aboriginal numbers



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Traditionally, First Nations Australians relied on story-telling and using the spoken language rather than writing, so Aboriginal people used words rather than symbols to describe numbers. Different regions had their own language and different names for numbers.

The **Belyando River people** of central Queensland use these words to name their numbers:

1 = wogin  2 = booleroo

3 = booleroo wogin  4 = booleroo booleroo

The **Kamilaroi people** live in northern New South Wales, including the regions surrounding Moree and Tamworth. They use 3 words to name their numbers.

1 = mal  2 = bularr  3 = guliba

4 = bularr bularr  5 = bularr guliba  6 = guliba guliba

Body parts such as fingers, arms, shoulders, neck and head are also used by some First Nations peoples to represent numbers when counting.

**1** How did the Belyando River people form words for the numbers 3 and 4?

**2** How did the Kamilaroi people form words for 4, 5 and 6?

**3** Write the answer to each expression, using the correct Aboriginal words:

**a** wogin + booleroo wogin  **b** guliba × bularr

**c** bularr + bularr + mal  **d** booleroo × booleroo

**e** guliba guliba – guliba  **f** bularr bularr – mal

**4** Find out more about how body parts are used to represent numbers by some First Nations Australians.

# Powers and index notation 3.07

The use of powers allows us to write repeated multiplication in a shorter way.

$4^2 = 4 \times 4$ '4 squared' or '4 to the power of 2'

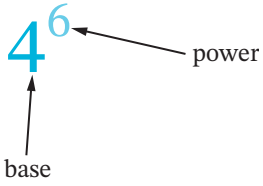
$4^3 = 4 \times 4 \times 4$ '4 cubed' or '4 to the power of 3'

$4^4 = 4 \times 4 \times 4 \times 4$ '4 to the power of 4'

$4^5 = 4 \times 4 \times 4 \times 4 \times 4$ '4 to the power of 5'

$4^6 = 4 \times 4 \times 4 \times 4 \times 4 \times 4$ '4 to the power of 6'

In $4^6$, the number 4 is called the **base** and is the number that is repeated in the multiplication. The small raised number 6 is called the **power** or **index**.




Video
The biggest number ever

3.07

## Example 12

Write each expression using **index notation**.

**a** $6 \times 6 \times 6 \times 6 \times 6$ **b** $7 \times 7 \times 7 \times (-2) \times (-2) \times (-2) \times (-2)$

### Solution

**a** $6 \times 6 \times 6 \times 6 \times 6 = 6^5$

**b** $7 \times 7 \times 7 \times (-2) \times (-2) \times (-2) \times (-2) = 7^3 \times (-2)^4$

## Example 13

Evaluate each expression.

**a** $11^2$ **b** $(-3)^3$ **c** $2^6$

### Solution

**a** $11^2 = 11 \times 11$
$= 121$

**b** $(-3)^3 = (-3) \times (-3) \times (-3)$
$= -27$

**c** $2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2$
$= 64$

**EXERCISE 3.07** ANSWERS ON P. 568

## Powers and index notation U F R C

**1** Write each expression using index notation. **c**

EXAMPLE 12

**a** $8 \times 8 \times 8 \times 8$ **b** $7 \times 7 \times 7 \times 7 \times 7 \times 7$

**c** $-5 \times (-5) \times (-5)$ **d** $1 \times 1 \times 1 \times 1 \times 1$

**e** $13 \times 13 \times 13 \times 13$ **f** $10$

**g** $(-4) \times (-4) \times 9 \times 9 \times 9$ **h** $2 \times 2 \times 12 \times 12$

**i** $3 \times 3 \times 11 \times 11 \times 11 \times 11$ **j** $15 \times (-6) \times (-6) \times 15$

**k** $2 \times 8 \times 2 \times 8 \times 2$ **l** $10 \times 10 \times 20$

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**2** Write each expression in expanded form (for example, $2^4 = 2 \times 2 \times 2 \times 2$). C

**a** $3^5$ **b** $5^2$ **c** $7^4$

**d** $(-6)^3$ **e** $2^4 \times 4^2$ **f** $12^3 \times 11^6$

EXAMPLE 13

**3** Evaluate each expression.

**a** $4^2$ **b** $5^3$ **c** $2^5$ **d** $10^6$

**e** $3^5$ **f** $7^4$ **g** $1^3$ **h** $(-2)^4$

**i** $3^3$ **j** $(-1)^2$ **k** $(-9)^5$ **l** $4^6$

Worksheet
Big numbers

**4** Copy and complete this table of powers of 10. R C

| Power of 10 | Number | Name |
|---|---|---|
| $10^1$ | 10 | ten |
| $10^2$ | 100 | hundred |
| $10^3$ | 1000 | |
| $10^6$ | 1 000 000 | |
| $10^9$ | | billion |
| $10^{12}$ | 1 000 000 000 000 | |
| $10^{15}$ | | quadrillion |
| | 1 000 000 000 000 000 000 | quintillion |
| $10^{21}$ | | sextillion |
| $10^{24}$ | | septillion |
| | 1 000 000 000 000 000 000 000 000 000 | octillion |
| $10^{30}$ | | nonillion |
| $10^{33}$ | 1 000 000 000 000 000 000 000 000 000 000 | decillion |

## Technology

### Powers

**1** Set up a spreadsheet to calculate powers by first entering the information shown below.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Number: | 2 | 3 | 5 | 7 |
| 2 | 1 | | | | |
| 3 | 2 | | | | |
| 4 | 3 | | | | |
| 5 | 4 | | | | |
| 6 | 5 | | | | |
| 7 | 6 | | | | |
| 8 | 7 | | | | |
| 9 | 8 | | | | |
| 10 | 9 | | | | |
| 11 | 10 | | | | |
| 12 | 11 | | | | |
| 13 | 12 | | | | |

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2 In cell B2, enter **=$B$1^A2** to calculate $2^1$, where ^ in a spreadsheet formula means 'to the power of'. We also use **$B$1** instead of **B1** because when we copy this formula, we want it to **always** refer to the **2** in cell B1, not another cell. This is called **absolute cell referencing**. We use this technique to maintain a particular value in a cell without changing it when writing a formula. However, we use **A2** instead of **$A$2** because when we copy this formula, we want it to refer to the different powers shown in column A to calculate $2^1$, $2^2$, $2^3$, and so on. This is called **relative cell referencing**.

3 Click on cell B2 and Fill Down to cell B13. Your spreadsheet will now show the first 12 powers of 2, up to $2^{12} = 4096$.

4 Enter similar formulas for cells C2, D2 and E2, and **Fill Down** for columns C, D and E to show the first 12 powers of 3, 5 and 7.

5 If a number is too long to show in a cell, it will either look something like **2·44E + 08** or **########**. For cells that show something like **2·44E + 08**, right-click, select **Format cells**, **Number** and **0** for decimal places. **########** means the column is not wide enough to show all digits of the number. You need to widen the column until you can see all digits.

## Did you know?

### Googol vs Google

The number $10^{100}$, the **googol,** is 1 followed by one hundred zeros. The name 'googol' was created by the 9-year-old nephew of American mathematician Dr Edward Kasner. The number $10^{\text{googol}}$, that is 1 followed by a googol zeros, is called the **googolplex**. The googol is a very big number but it is rarely used for practical purposes. Even the number of particles in the observable universe, estimated at being between $10^{72}$ and $10^{87}$, is less than a googol!

The Internet search engine Google was named after the googol, to reflect the huge size of the World Wide Web. It was created in 1996 by two Stanford University students, Larry Page and Sergey Brin. They even named Google's global headquarters in California the Googleplex. Google is a powerful search engine because it can find information from over a trillion (a million million) web pages in less than 1 second.

**How many googols are there in a googolplex?**




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# 3.08 Square root and cube root

**Skillsheet**
Square roots and cube roots

**Worksheet**
Powers and roots

**Puzzle**
Square root Snap

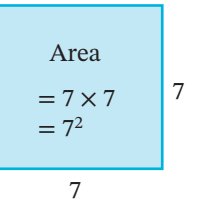
## Square and square root

**Squaring** a number means raising it to the **power of 2**.

For example, 'squaring 7' or 'the square of 7' means $7^2 = 7 \times 7 = 49$, read '7 squared'. Also, 49 is called a **square number** or **perfect square** because it is the square of a whole number.

It is called '7 squared' because it is the area of a square of side 7 units.

The opposite of squaring is the **square root** (symbol $\sqrt{\ }$).




### Square root

The **square root** $\sqrt{\ }$ of a number is the positive value which, when squared, gives that number.

For example, $\sqrt{49} = 7$, read 'the square root of 49', because $7^2 = 7 \times 7 = 49$.

### Example 14

Evaluate:

**a** the square root of 64 **b** $\sqrt{9}$ **c** $\sqrt{25}$

#### Solution

**a** The square root of 64 $= \sqrt{64} = 8$ because $8^2 = 8 \times 8 = 64$

**b** $\sqrt{9} = 3$ because $3^2 = 3 \times 3 = 9$

**c** $\sqrt{25} = 5$ because $5^2 = 5 \times 5 = 25$

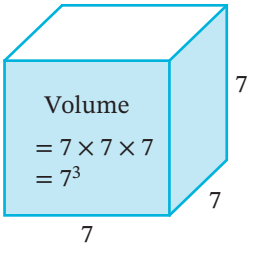
## Cube and cube root

**Cubing** a number means raising it to the **power of 3**.

For example, 'cubing 7' or 'the cube of 7' means $7^3 = 7 \times 7 \times 7 = 343$, read '7 cubed'. Also, 343 is called a **cube number** or **perfect cube** because it is the cube of a whole number.

It is called '7 cubed' because it is the volume of a cube of side 7 units.

The opposite of cubing is the **cube root** (symbol $\sqrt[3]{\ }$).




### Cube root

The **cube root** $\sqrt[3]{\ }$ of a number is the value which, when cubed, gives that number.

For example, $\sqrt[3]{343} = 7$, read 'the cube root of 343', because $7^3 = 7 \times 7 \times 7 = 343$.


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## Example 15

Evaluate:

**a** the cube root of 125 **b** $\sqrt[3]{8}$ **c** $\sqrt[3]{-64}$

### Solution

**a** The cube root of 125 $= \sqrt[3]{125} = 5$ because $5^3 = 5 \times 5 \times 5 = 125$

**b** $\sqrt[3]{8} = 2$ because $2^3 = 2 \times 2 \times 2 = 8$

**c** $\sqrt[3]{-64} = -4$ because $(-4)^3 = -4 \times (-4) \times (-4) = -64$

Note: The square root of a number is positive but a cube root can be positive or negative. However, the square root and cube root of 0 are both 0.

## Example 16

Estimate the value of $\sqrt{40}$.

### Solution

There is no *exact* answer for the square root of 40, because there isn't a number which, if squared, equals 40 exactly. However, we can find a decimal whose square is close to 40.

Noting that:

$5^2 = 25$

$6^2 = 36$

$7^2 = 49$

we can tell that $\sqrt{40}$ must lie somewhere between 6 and 7, and closer to 6 because 40 is closer to 36 than 49.

So we can estimate that $\sqrt{40} \approx 6.3$.

(In fact, $6.3^2 = 39.69$).

**EXERCISE 3.08** ANSWERS ON P. 569

## Square root and cube root UFRC

**1** Evaluate each square number. C

**a** $9^2$ **b** $17^2$ **c** $10^2$

**2** The $x^2$ key on a calculator can be used to square a number. For example, to find $9^2$, press 9 $x^2$ =. Use your calculator to evaluate each square number.

**a** $7^2$ **b** $13^2$ **c** $18^2$

**d** $1^2$ **e** $5^2$ **f** $20^2$

**g** $16^2$ **h** $34^2$ **i** $15^2$

**j** $(-3)^2$ **k** $(-7)^2$ **l** $(-13)^2$

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EXAMPLE 14

**3** Use your answers from question **2** to find: R

**a** $\sqrt{400}$ **b** $\sqrt{225}$ **c** $\sqrt{324}$

**d** $\sqrt{256}$ **e** $\sqrt{169}$ **f** $\sqrt{1156}$

**4** Copy and complete this table.

| Number, $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Number squared, $x^2$ | | | | | | | | | | | | |

**5** Which number below is a square number? Select the correct answer **A**, **B**, **C** or **D**.

**A** 32 **B** 36 **C** 40 **D** 45

**6** Use your answers from the table in question **4** to find: R

**a** the square root of 100 **b** $\sqrt{36}$ **c** $\sqrt{121}$

**d** $\sqrt{16}$ **e** $\sqrt{1}$ **f** the square root of 81

**g** $\sqrt{64}$ **h** the square root of 144 **i** $\sqrt{25}$

**7** The $\sqrt{\ }$ key on a calculator can be used to find the square root of a number. For example, to find $\sqrt{64}$, press $\sqrt{\ }$ 64 =. Use your calculator to find each square root.

**a** $\sqrt{49}$ **b** $\sqrt{81}$ **c** $\sqrt{625}$

**d** $\sqrt{484}$ **e** $\sqrt{1764}$ **f** $\sqrt{361}$

**g** $\sqrt{900}$ **h** $\sqrt{784}$ **i** $\sqrt{256}$

**j** $\sqrt{196}$ **k** $\sqrt{400}$ **l** $\sqrt{3136}$

**8** There are 2 numbers whose square root is itself, that is, $\sqrt{x} = x$.
What are the 2 numbers? R

**9** Evaluate each cube number. C

**a** $4^3$ **b** $11^3$ **c** $1^3$

**10** The $x^3$ key on a calculator can be used to cube a number. For example, to find $4^3$, press 4 $x^3$ =. Use your calculator to find each cube number.

**a** $7^3$ **b** $16^3$ **c** $5^3$

**d** $10^3$ **e** $20^3$ **f** $17^3$

**g** $35^3$ **h** $(-8)^3$ **i** $(-5)^3$

**11** Use your answers from question **10** to find: R

**a** $\sqrt[3]{-512}$ **b** $\sqrt[3]{8000}$ **c** $\sqrt[3]{42\,875}$

**d** $\sqrt[3]{-125}$ **e** $\sqrt[3]{4913}$ **f** $\sqrt[3]{4096}$

**12** Copy and complete this table.

| Number, $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Number cubed, $x^3$ | | | | | | | | | | | | |

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**13** Which number below is a cube number? Select **A**, **B**, **C** or **D**.

**A** 192 **B** 512 **C** 625 **D** 800

EXAMPLE 15

**14** Use your answers from question **12** to find: R

**a** the cube root of 512 **b** $\sqrt[3]{729}$ **c** $\sqrt[3]{125}$

**d** $\sqrt[3]{1}$ **e** $\sqrt[3]{27}$ **f** the cube root of 1728

**g** $\sqrt[3]{64}$ **h** the cube root of 1331 **i** $\sqrt[3]{1000}$

**15** The $\sqrt[3]{\ }$ key on a calculator can be used to find the cube root of a number. For example, to find $\sqrt[3]{64}$, press $\sqrt[3]{\ }$ 64 =. Use your calculator to find each cube root.

**a** $\sqrt[3]{4096}$ **b** $\sqrt[3]{2744}$ **c** $\sqrt[3]{10\,648}$

**d** $\sqrt[3]{64\,000}$ **e** $\sqrt[3]{19\,683}$ **f** $\sqrt[3]{9261}$

**16** Use your answers from the table in question **4** to determine between which 2 consecutive whole numbers $\sqrt{27}$ must lie. Select **A**, **B**, **C** or **D**. R

EXAMPLE 16

**A** 26 and 28 **B** 13 and 14 **C** 4 and 5 **D** 5 and 6

**17** Determine between which 2 consecutive numbers $\sqrt{15}$ must lie. Select **A**, **B**, **C** or **D**. R

**A** 7 and 8 **B** 3 and 4 **C** 196 and 197 **D** 4 and 5

**18** Estimate the value of each root, then use your calculator to check. R

**a** $\sqrt{50}$ **b** $\sqrt{32}$ **c** $\sqrt{96}$

**d** $\sqrt[3]{71}$ **e** $\sqrt[3]{900}$ **f** $\sqrt[3]{184}$

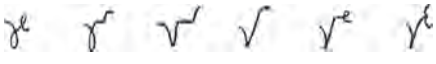
3.08

## Did you know?

### The radical symbol

The formal name for the square root symbol ($\sqrt{\ }$) is the **radical symbol.** This symbol was created by the German mathematician Christoff Rudolff in 1525, but it was not until the end of the 17th century that the symbol was widely accepted. Before this, the most commonly-used symbol for square root was Rx. It was first used in 1220 and was an abbreviation for the word **radix**, which means 'root' in Latin. One theory says that Rudolff invented the symbol $\sqrt{\ }$ to look like the letter 'r'.

Other early symbols for square root were



**The symbols on the right all represent the same thing.**

**What do you think they stand for?**

Foundation Standard Complex

Quiz
Mental skills 3B

## Mental skills 3B: Maths without calculators ANSWERS ON P. 569

### Dividing by a multiple of 10

Place value allows us to remove 0s from the end of a number when we divide by a power of 10.

**1** Study each example.

**a** $2000 \div 10 = 200\cancel{0} \div 1\cancel{0} = 200$

**b** $1800 \div 100 = 18\cancel{00} \div 1\cancel{00} = 18$

**c** $37\,000 \div 100 = 37\,0\cancel{00} \div 1\cancel{00} = 370$

**d** $5\,000\,000 \div 1000 = 5\,000\,\cancel{000} \div 1\cancel{000} = 5000$

**e** $6000 \div 200 = 60\cancel{00} \div 1\cancel{00} \div 2 = 60 \div 2 = 30$

**f** $350 \div 70 = 35\cancel{0} \div 1\cancel{0} \div 7 = 35 \div 7 = 5$

**g** $2800 \div 40 = 280\cancel{0} \div 1\cancel{0} \div 4 = 280 \div 4 = 70$

**h** $40\,000 \div 5000 = 40\,\cancel{000} \div 1\cancel{000} \div 5 = 40 \div 5 = 8$

**2** Now evaluate each quotient.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | $200 \div 10$ | **b** | $6000 \div 100$ | **c** | $45\,000 \div 100$ | **d** | $30\,000 \div 1000$ |
| **e** | $1900 \div 10$ | **f** | $2600 \div 100$ | **g** | $530 \div 10$ | **h** | $720\,000 \div 1000$ |
| **i** | $180 \div 30$ | **j** | $300 \div 50$ | **k** | $1600 \div 400$ | **l** | $45\,000 \div 5000$ |
| **m** | $4200 \div 60$ | **n** | $21\,000 \div 700$ | **o** | $44\,000 \div 2000$ | **p** | $1600 \div 200$ |
| **q** | $24\,000 \div 600$ | **r** | $15\,000 \div 3000$ | **s** | $64\,000 \div 80$ | **t** | $5400 \div 900$ |

# 3.09 Prime and composite numbers

Skillsheet
Prime and composite numbers

Worksheet
Sieve of Eratosthenes

The **factors** of a number are those whole numbers that divide exactly into it (without remainder). For example:

- The factors of 16 are 1, 2, 4, 8 and 16
- The factors of 7 are 1 and 7

A number is **prime** if it has only **2** factors: 1 and itself. Some examples of prime numbers are 7, 3, 19, 2 and 11.

A number is **composite** if it has **more than 2** factors. Some examples of composite numbers are 16, 10, 9, 15 and 4.

Video
The prime number code

## Prime and composite numbers

- A **prime number** has only 2 factors: 1 and itself
- A **composite number** has more than 2 factors

*Note*: The number 1 has only one factor so 1 is neither prime nor composite.

**EXERCISE 3.09** ANSWERS ON P. 569

## Prime and composite numbers U F R C

You can use the divisibility tests to help find the factors of larger numbers.

**1** List all the factors of each number.

| | | | |
|---|---|---|---|
| **a** 17 | **b** 21 | **c** 24 | **d** 11 |
| **e** 35 | **f** 4 | **g** 18 | **h** 23 |
| **i** 25 | **j** 9 | **k** 3 | **l** 19 |

**2** List all the numbers from question **1** that are:

**a** prime **b** composite

**3** Which one of these numbers is not a factor of 45? Select the correct answer **A**, **B**, **C** or **D**.

**A** 9 **B** 5 **C** 7 **D** 3

**4** The ancient Greek mathematician, Eratosthenes, discovered an easy way to sort out the prime numbers from a list of numbers. It is called the Sieve of Eratosthenes (pronounced 'Siv of Era-tos-the-nees'), and works by crossing out multiples of numbers (the composite numbers). R

**a** Copy the grid below or print out the Worksheet 'Sieve of Eratosthenes'.

| | | | | | |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 7 | 8 | 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 | 17 | 18 |
| 19 | 20 | 21 | 22 | 23 | 24 |
| 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 |
| 37 | 38 | 39 | 40 | 41 | 42 |
| 43 | 44 | 45 | 46 | 47 | 48 |
| 49 | 50 | 51 | 52 | 53 | 54 |
| 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 |
| 67 | 68 | 69 | 70 | 71 | 72 |
| 73 | 74 | 75 | 76 | 77 | 78 |
| 79 | 80 | 81 | 82 | 83 | 84 |
| 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 |
| 97 | 98 | 99 | 100 | 101 | 102 |
| 103 | 104 | 105 | 106 | 107 | 108 |
| 109 | 110 | 111 | 112 | 113 | 114 |
| 115 | 116 | 117 | 118 | 119 | 120 |

**b** Cross out 1. It is neither prime nor composite.

**c** Except for 2, colour all the multiples of 2 red and notice the pattern.

**d** Except for 3, colour all the multiples of 3 green and notice the pattern.

**e** Continue to colour multiples of other numbers with different colours, until there are no more multiples.

**f** What do you notice about the 30 numbers left that are not coloured?

3.09

**Worksheet** Sieve of Eratosthenes

Foundation Standard Complex

**5** Sort the following numbers into primes and composites.

10 2064 101 129 33 77 967 504 999 17 27 41 71 59 472 62

**6** List all the:

**a** prime numbers between 36 and 50

**b** composite numbers between 65 and 80

**c** prime numbers less than 20

**d** composite numbers between 30 and 47

**7** Look up other meanings for the word 'composite'. Suggest why this word is used the way it is in mathematics. C

**8** Which number below is divisible only by prime numbers, itself and 1? Select **A**, **B**, **C** or **D**.

**A** 12 **B** 14 **C** 16 **D** 18

**9a** Find a number that has an odd number of factors.

**b** What type of numbers have an odd number of factors? R C

# 3.10 Factor trees

**Worksheet**
Factor trees

**Skillsheet**
Prime factors by repeated division

**Video**
Factor trees

Every number can be written as a product of its prime factors. The prime factors can be found by using a **factor tree**.

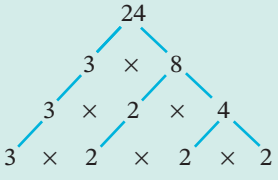
## Example 17

Write 24 as a product of its prime factors.

### Solution

Draw a factor tree for 24.

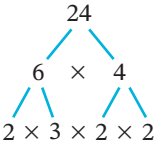
- 8 is composite, so write 8 as a product of 2 factors.
- 3 is prime, so keep writing 3 along the branches.




So 24 as a product of prime factors is:

$24 = 2 \times 2 \times 2 \times 3$.

*Note*: It is possible to draw different factor trees using different factors for the same number, but the final list of prime factors should still be the same. Here is another factor tree for 24:




$24 = 2 \times 2 \times 2 \times 3$

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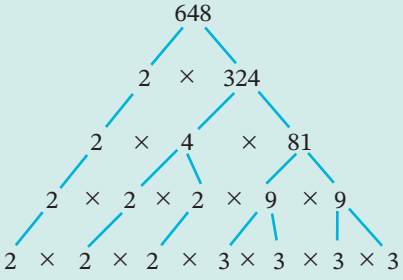
## Example 18

Write 648 as a product of its prime factors, using index notation.

### Solution

Draw a factor tree for 648.

- 2 is prime, so keep writing 2 along the branches.
- 324 is composite, so write 324 as a product of 2 factors, 4 and 81.
- Keep going until there are no composite factors left.




So 648 as a product of prime factors is:

$$648 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3$$
$$= 2^3 \times 3^4$$

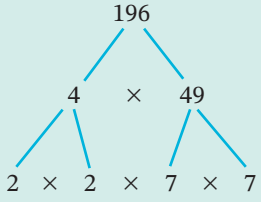
## Example 19

Use a factor tree to find the value of:

**a** $\sqrt{196}$ **b** $\sqrt[3]{216}$

### Solution

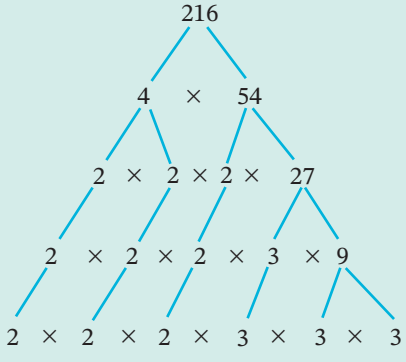
**a** Draw a factor tree for 196.




So $196 = 2 \times 2 \times 7 \times 7$

$$\therefore \sqrt{196} = \sqrt{2 \times 2 \times 7 \times 7}$$
$$= \sqrt{2 \times 2} \times \sqrt{7 \times 7}$$
$$= 2 \times 7$$
$$= 14$$

**b** Draw a factor tree for 216.




So $216 = 2 \times 2 \times 2 \times 3 \times 3 \times 3$

$$\therefore \sqrt[3]{216} = \sqrt[3]{2 \times 2 \times 2 \times 3 \times 3 \times 3}$$
$$= \sqrt[3]{2 \times 2 \times 2} \times \sqrt[3]{3 \times 3 \times 3}$$
$$= 2 \times 3$$
$$= 6$$



**Video**
Using factor trees to find square roots

**EXERCISE 3.10** ANSWERS ON P. 569

## Factor trees U F R

EXAMPLE 17

**1** Use a factor tree to write each number as a product of its prime factors.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | 88 | **b** | 63 | **c** | 45 | **d** | 51 |
| **e** | 132 | **f** | 270 | **g** | 396 | **h** | 218 |
| **i** | 630 | **j** | 520 | **k** | 275 | **l** | 342 |

**2** What is 1260 expressed as a product of its prime factors?
Select the correct answer **A**, **B**, **C** or **D**.

| | | | |
|---|---|---|---|
| **A** | $2 \times 3 \times 3 \times 3 \times 5 \times 7$ | **B** | $2 \times 2 \times 2 \times 3 \times 5 \times 7$ |
| **C** | $2 \times 2 \times 3 \times 3 \times 3 \times 7$ | **D** | $2 \times 2 \times 3 \times 3 \times 5 \times 7$ |

**3** Draw 2 different factor trees for 280 and show that both give the same prime factors.

EXAMPLE 18

**4** Use a factor tree to write each number as a product of its prime factors, in index notation.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | 48 | **b** | 200 | **c** | 460 | **d** | 712 |
| **e** | 98 | **f** | 144 | **g** | 325 | **h** | 135 |

EXAMPLE 19

**5** 484 is a perfect square. R

**a** Use a factor tree to write 484 as a product of its prime factors.

**b** Note that because 484 is a square number, the prime factors are in pairs.
Use the factor tree to find $\sqrt{484}$.

**6** Use a factor tree to evaluate each root. R

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | $\sqrt{625}$ | **b** | $\sqrt{900}$ | **c** | $\sqrt{225}$ | **d** | $\sqrt{784}$ |
| **e** | $\sqrt[3]{9261}$ | **f** | $\sqrt[3]{3375}$ | **g** | $\sqrt[3]{1728}$ | **h** | $\sqrt[3]{4096}$ |

## Technology

### Prime factors calculator

Search the Internet for an online prime factors calculator. This is a calculator that will take a number that you type in and write it as a product of its prime factors, perhaps even drawing its factor tree. Use it to show that $648 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 = 2^3 \times 3^4$, then use it to check some of your answers from the previous exercise.

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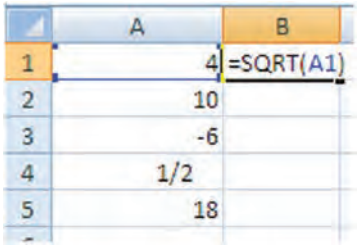
# Technology

## Spreadsheet formulas

Remember that a spreadsheet is like a calculator. We use special symbols to make calculations and a spreadsheet formula always begins with an equals (=) sign.

| Formula | Meaning |
|---|---|
| =A1+A2+A3 or =sum(A1:A3) | **add** the values in cells A1, A2 and A3 |
| =A5-A4 | **subtract** the value in cell A4 from the value in cell A5 |
| =A1*A3 | **multiply** together the values in cells A1 and A3 (* is used instead of ×) |
| =A1/A2 | **divide** the value in cell A1 by the value in cell A2 (/ is used instead of ÷) |
| =A2^2 | **square** the value in cell A2 (^2 is used instead of $(A2)^2$) |
| =A5^3 | **cube** the value in cell A5, that is, $(A5)^3$ |
| =sqrt(A3) | find the **square root** of the value in cell A3, that is, $\sqrt{A3}$ |
| =average(A1:A5) | find the **average** of all values from cells A1 to A5 |
| =max(A1:A8) | find the **maximum** (**highest**) of all values from cells A1 to A8 |
| =min(A1:A8) | find the **minimum** (**lowest**) of all values from cells A1 to A8 |

**1** Open a new spreadsheet and enter the 5 numbers shown in column A. To enter the fraction **1/2** in cell A4, right-click on B4 to select **Format Cells**, then choose **Fraction** and select **Up to three digits**.




**2** In cell B1, enter the formula **=sqrt(A1)** to calculate the square root of the value in cell A1. You should get the answer 2, because $\sqrt{4} = 2$.

**3** For the rest of the cells in column B, enter the appropriate formula for each expression shown below and check that the spreadsheet calculates the correct answer. Remember to start each formula with an '=' sign.

**a** at B2, A2 ÷ A1

**b** at B3, $(A3)^2$

**c** at B4, $(A3)^3$

**d** at B5, A1 × A2 + A3 × A4

**e** at B6, A5 – A4

**f** at B7, the sum of A1 to A5

**g** at B8, the product of A1 to A5

**h** at B9, $(A4)^2$

**i** at B10, $(A5)^3$

**j** at B11, $\sqrt{\frac{A5}{A4}}$

4 For the following cells in column C, enter the appropriate formula for each expression and check that the spreadsheet calculates the correct answer.

   a at C1, the average of column A

   b at C2, the maximum value of column A

   c at C3, the minimum value of column A

   d at C4, $(A1 + A5)^2$

   e at C5, the average of A1, A2 and A3

   f at C6, $\sqrt{A2 - A3}$

5 Now enter **new** values in cells A1 to A5 (including a new fraction for A4) and notice the new answers calculated in columns B and C.

6 Create your own values and formulas. Write at least five formulas and use up to five different values. Enter the new values in column D and the new formulas in column E of your spreadsheet.

# 3.11 Highest common factor

**Presentation**
Highest common factors

## Highest common factor

The **highest common factor** (HCF) or **greatest common divisor** (GCD) of 2 (or more) numbers is the largest number that is a factor of **both** (**or all**) of these numbers.

**Divisor** is just another name for **factor**.

**Video**
Highest common factor

## Example 20

Find the highest common factor (HCF) of:

**a** 24 and 30 **b** 40 and 15

### Solution

**a** Factors of 24 = **1**, **2**, **3**, 4, **6**, 8, 12, 24

Factors of 30 = **1**, **2**, **3**, 5, **6**, 10, 15, 30

The common factors are 1, 2, 3, 6

The highest common factor is 6.

**b** Factors of 40 = **1**, 2, 4, **5**, 8, 10, 20, 40

Factors of 15 = **1**, 3, **5**, 15

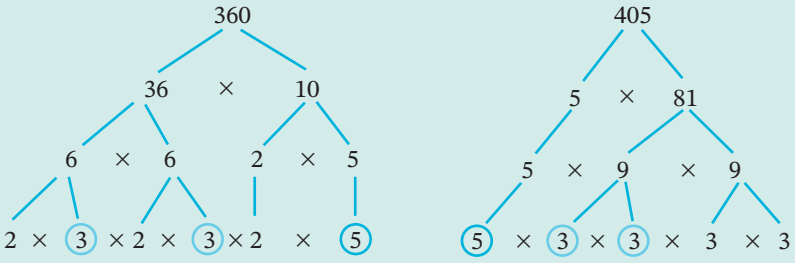
The highest common factor is 5.

# Using prime factors to find the HCF

## Example 21

Use factor trees to find the HCF of 360 and 405.

### Solution




- Draw factor trees for 360 and 405
- Circle common prime factors: 3, 3 and 5
- Multiply 3, 3 and 5 to calculate the HCF

HCF $= 3 \times 3 \times 5$

$= 45$ 45 is the highest factor of both 360 and 405

## Using prime factors to find the HCF

To find the highest common factor of 2 numbers using their prime factors:

**1** Circle common prime factors

**2** Multiply them together

Video
HCF and factor trees

**EXERCISE 3.11** ANSWERS ON P. 570

## Highest common factor UFRC

**1** Which one of these numbers is **not** a factor of 36? Select the correct answer **A**, **B**, **C** or **D**.

**A** 4 **B** 6 **C** 8 **D** 9

**2** Find all common factors for each pair of numbers.

**a** 27, 45 **b** 60, 32 **c** 75, 45 **d** 18, 28

**3** Find the highest common factor of each pair of numbers.

EXAMPLE 20

**a** 32, 28 **b** 9, 6 **c** 6, 14 **d** 8, 12

**e** 14, 70 **f** 44, 64 **g** 10, 15 **h** 12, 6

**i** 75, 125 **j** 56, 40 **k** 60, 90 **l** 39, 26

**m** 132, 60 **n** 36, 84 **o** 27, 63 **p** 350, 210

**4** On a spreadsheet, the highest common factor can be found using the function **=gcd()**.

**a** Enter **=gcd(27, 45)** into a spreadsheet cell to find the HCF of 27 and 45.

**b** Check your answers to question **3** using a spreadsheet.

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EXAMPLE 21

**5** Use factor trees to find the highest common factor of each pair of numbers. R

**a** 324, 486 **b** 144, 60 **c** 35, 210 **d** 1404, 1900

**6** Explain why the 'factor trees' method for finding the HCF works, using 324 and 486 from question **5a** as an example. R C

## Investigation

### Common factor puzzle

**1** Copy this grid, and try to move from the 200 square in the top left corner to the 100 square at the bottom right corner. You can move one step horizontally or vertically (but not diagonally), but only if both numbers have a common factor (not 1). For example, you can move from 65 to 91 (common factor 13), but not to 96 (no common factor).

| | | | | | | |
|---|---|---|---|---|---|---|
| 200 | 80 | 65 | 91 | 143 | 156 | 195 |
| 175 | 32 | 96 | 71 | 110 | 77 | 121 |
| 35 | 28 | 15 | 209 | 87 | 90 | 21 |
| 39 | 169 | 117 | 95 | 57 | 37 | 81 |
| 63 | 11 | 29 | 72 | 76 | 75 | 51 |
| 14 | 98 | 56 | 132 | 48 | 78 | 85 |
| 105 | 45 | 44 | 187 | 112 | 221 | 100 |

**2** Now try to move from the 105 square in the bottom left corner to the 195 square at the top right corner, using the same rules.

**3** Choose different starting and finishing positions. Do they all have connecting factor paths?

## Investigation

### HCF by repeated division

The highest common factor can be found by repeated division with prime factors.
For example, to find the HCF of 24 and 60:

- both numbers are even so divide by 2
- both numbers are even so divide by 2 again
- both numbers are divisible by 3 so divide by 3
- Since it is not possible to divide any more, stop.
- Multiply the common prime factors to find the HCF.
- HCF of 24 and 60 = $2 \times 2 \times 3 = 12$

$$\begin{array}{rr} \rightarrow 2)\mathbf{24} & 2)\mathbf{60} \\ \rightarrow 2)\mathbf{12} & 2)\mathbf{30} \\ \rightarrow 3)\ \mathbf{6} & 3)\mathbf{15} \\ 2 & 5 \end{array}$$

Use this method to find the HCF of each set of numbers.

**a** 45, 120 **b** 18, 48 **c** 20, 28

**d** 96, 144 **e** 16, 48 **f** 15, 21, 45

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# Extension: Lowest common multiple 3.12

EXTENSION



- The multiples of 3 are 3, 6, 9, 12, 15, 18, 21, 24, ...
- The multiples of 7 are 7, 14, 21, 28, 35, 42, 49, 56, ...

The **multiples** of a number are found by multiplying that number by the whole numbers 1, 2, 3, 4, ...

## Lowest common multiple

The **lowest common multiple** (LCM) of 2 (or more) numbers is the smallest number that is a multiple of **both** (**or all**) of these numbers.

## Example 22

Find the lowest common multiple (LCM) of:

**a** 3 and 4 **b** 6 and 9

### Solution

**a** Multiples of 4 are 4, 8, **12**, 16, 20, **24**, 28, 32, ...
Multiples of 3 are 3, 6, 9, **12**, 15, 18, 21, **24**, ...
The common multiples are **12**, **24**, ...
The lowest common multiple of 3 and 4 is **12**.

**b** Multiples of 6 are 6, 12, **18**, 24, 30, **36**, 42, 48, **54**, ...
Multiples of 9 are 9, **18**, 27, **36**, 45, **54**, 63, 72, 81, ...
The lowest common multiple of 6 and 9 is **18**.

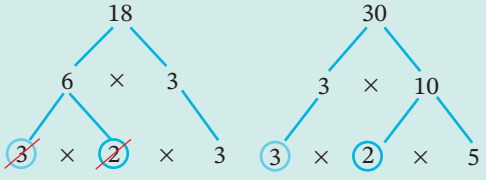
# Using prime factors to find the LCM

## Example 23

Use factor trees to find the LCM of 18 and 30.

### Solution

- Draw factor trees for 18 and 30.
- Circle common prime factors: 3 and 2.
- Cross out one of the 3s and one of the 2s.
- Multiply the remaining factors to calculate the LCM.




LCM of 18 and 30 = 3 × 3 × 2 × 5
= 90

EXTENSION

Notice that a common multiple of 18 and 30 is $18 \times 30 = 540$, but to find the lowest common multiple, you can divide that product by their HCF $= 3 \times 2 = 6$, giving $540 \div 6 = 90$.
This means that:

$$\text{LCM of } a \text{ and } b = \frac{a \times b}{\text{HCF of } a \text{ and } b}.$$

## Using prime factors to find the LCM

To find the lowest common multiple of 2 numbers using their prime factors:

**1** Circle common prime factors

**2** Cross out **one** of each pair of common factors

**3** Multiply the remaining factors

### EXERCISE 3.12 ANSWERS ON P. 570

## Lowest common multiple U F R C

**1** Write the first 5 multiples of 7.

**2** Which number is NOT a multiple of 8? Select the correct answer **A**, **B**, **C** or **D**.

**A** 24 **B** 36 **C** 56 **D** 72

**3** List the multiples of 4 between 13 and 45.

**4** How many multiples of 13 are less than 220?

EXAMPLE 22

**5** Find the lowest common multiple of each set of numbers.

**a** 3, 5 **b** 6, 7 **c** 4, 6 **d** 15, 10

**e** 5, 8 **f** 4, 10 **g** 10, 5 **h** 2, 8

**i** 9, 6 **j** 3, 7 **k** 3, 4, 5 **l** 4, 12, 10

**6** On a spreadsheet, the lowest common multiple can be found using the function **=lcm()**.

**a** Enter **=lcm(8, 12)** into a spreadsheet cell to find the LCM of 8 and 12.

**b** Check your answers to question **5** using a spreadsheet.

**7** Why are some lowest common multiples easier to find? Write down your strategy for finding LCMs. R C

EXAMPLE 23

**8** Use factor trees to find the lowest common multiple of each pair of numbers. R

**a** 28, 15 **b** 16, 25 **c** 44, 20 **d** 18, 60

**9** Explain why the 'factor trees' method for finding the LCM works, using 28 and 15 from question **8a** as an example. R C

**10a** Find the HCF of 95 and 76.

**b** Use the formula (rule)

$$\text{LCM of } a \text{ and } b = \frac{a \times b}{\text{HCF of } a \text{ and } b}.$$

to find the LCM of 95 and 76.

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## Power plus ANSWERS ON P. 570



**Worksheet**
Perfect and amicable numbers



**1** For each set of numbers, evaluate and write in ascending order.

**a** $2^3, 3^2, 3^5, 5^3, 2^5, 5^2$

**b** $4^4, 7^3, 3^5, 8^2, 5^2, 6^3$

**c** $100^2, 11^4, 2^7, 3^4, 5^3$

**2** Evaluate each square number.

**a** $1^2$ **b** $11^2$ **c** $111^2$ **d** $1111^2$

**3** Based on the patterns in your answers to question **2**, evaluate each square number.

**a** $11\,111^2$ **b** $1\,111\,111^2$ **c** $111\,111\,111^2$ **d** $1\,111.1111^2$

**4** Find the square root of:

**a** $3 \times 3 \times 2 \times 2$ **b** $5 \times 5 \times 4 \times 4 \times 3 \times 3$

**5** Evaluate each square root without using a calculator.

**a** $\sqrt{9^2}$ **b** $\sqrt{3^4}$ **c** $\sqrt{5^4 \times 2^6}$

**d** $\sqrt[3]{2^9}$ **e** $\sqrt[3]{3^3 \times 5^6}$ **f** $\sqrt[4]{16}$

**6** Find the square root of each number without using a calculator.

**a** 2500 **b** 8100 **c** 10 000

**d** 1 000 000 **e** 1 210 000 **f** 100 000 000

**g** 640 000 **h** 176 400 **i** 10 000 000 000

**7** Two prime numbers that differ by 2 are called **twin primes**. For example, 11 and 13 are twin primes, but 23 and 29 are not. Find all 8 pairs of twin primes below 100.

# CHAPTER 3 REVIEW

**Quiz**
Language of maths 3

**Puzzle**
Number find-a-word

## Language of maths

| | | | |
|---|---|---|---|
| composite | cube | divisibility test | divisor |
| estimate | factor | factor tree | highest common factor (HCF) |
| index notation | power | prime | prime factor |
| round down | round up | square | square root |

**1** Which positive integer is neither prime nor composite?

**2** What notation is used as an abbreviation for repeated multiplication?

**3** What type of numbers have more than 2 factors?

**4** Describe the divisibility test for 3.

**5** Find a non-mathematical meaning of:

**a** prime **b** index

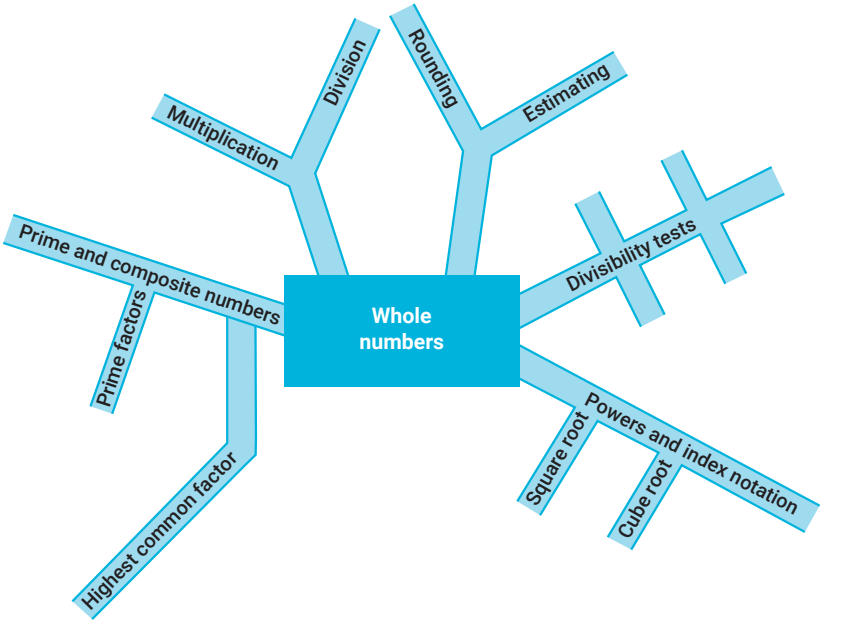
**6** What helps us to write a number as a product of its prime factors?

**Worksheet**
Mind map: Whole numbers

## Topic summary

- Using your own words, write down what you have learnt about whole numbers.
- What was your favourite part of this topic?
- What parts of this topic didn't you understand? Talk to your teacher or a friend about them.

Copy and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.





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# TEST YOURSELF 3

ANSWERS ON P. 570

Quiz
Test yourself 3



**1** Round 87 531 to the nearest: 3.01

**a** thousand **b** hundred **c** ten **d** ten thousand

**2** Estimate the value of each expression. 3.01

**a** $425 + 81 + 940 + 22$ **b** $864 + 250 - 417$

**c** $93 \times 19$ **d** $2402 \div 32$

**3** Evaluate each product. 3.02

**a** $12 \times 4$ **b** $34 \times 5$ **c** $33 \times 9$

**d** $25 \times 8$ **e** $126 \times 7$ **f** $565 \times 18$

**4** Evaluate each quotient. 3.03

**a** $2400 \div 10$ **b** $620 \div 5$ **c** $1296 \div 8$

**d** $528 \div 4$ **e** $144 \div 9$ **f** $847 \div 7$

**5** Test whether 1208 is divisible by: 3.04

**a** 2 **b** 3 **c** 4 **d** 5

**e** 6 **f** 8 **g** 9 **h** 10

**6** Which number is divisible by both 3 and 8? Select the correct answer **A**, **B**, **C** or **D**. 3.04

**A** 30 **B** 24 **C** 90 **D** 94

EXTENSION

**7** Evaluate each quotient. 3.05

**a** $3079 \div 12$ **b** $650 \div 25$ **c** $2976 \div 24$

**8** What does each Roman numeral mean? 3.06

**a** XXXVII **b** CCXL **c** XXIV

**9** Write each number as a Roman numeral. 3.06

**a** 125 **b** 28 **c** 74

**10** Write each expression using index notation. 3.07

**a** $3 \times 3 \times 3 \times 3$

**b** $6 \times 6 \times 6 \times 6 \times 6$

**c** $11 \times 11 \times 11 \times 11 \times 11 \times 11 \times 11$

**11** Evaluate each power. 3.07

**a** $6^2$ **b** $5^3$ **c** $10^5$ **d** $2^8$

**12** Evaluate each root. 3.08

**a** $\sqrt{49}$ **b** $\sqrt{144}$ **c** $\sqrt[3]{8000}$ **d** $\sqrt[3]{729}$

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3.09

**13** **a** List the prime numbers between 26 and 50.

**b** List the composite numbers between 45 and 70.

**c** List the prime numbers less than 30.

**d** List the composite numbers larger than 40 but less than 59.

3.10

**14** Use a factor tree to express each number as a product of its prime factors in index notation.

**a** 60 **b** 200 **c** 144

3.10

**15** Use a factor tree to evaluate each root.

**a** $\sqrt{256}$ **b** $\sqrt[3]{5832}$

3.11

**16** Find the highest common factor of each pair of numbers.

**a** 20 and 48 **b** 36 and 84

3.11

**17** Use factor trees to find the highest common factor of 243 and 162.

EXTENSION

3.12

**18** Find the lowest common multiple of each pair of numbers.

**a** 6 and 8 **b** 18 and 10

3.12

**19** Use factor trees to find the lowest common multiple of 25 and 16.

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# PRACTICE SET 1

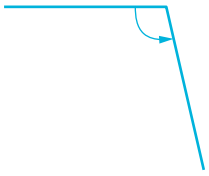
ANSWERS ON P. 570

**1** Evaluate each expression. 1.04-1.07

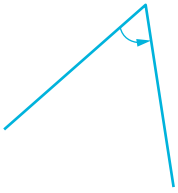
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | $-14 + 20$ | **b** | $11 - 16$ | **c** | $6 \times (-5)$ | **d** | $-48 \div (-8)$ |
| **e** | $22 + (-4)$ | **f** | $8 - (-9)$ | **g** | $-10 \times (-3)$ | **h** | $-42 \div 7$ |
| **i** | $3 - 5$ | **j** | $-2 + 8$ | **k** | $-1 - 6$ | **l** | $-3 + 5 - 8$ |
| **m** | $-5 \times (-3)$ | **n** | $6 \div (-2)$ | **o** | $-27 \div (-3)$ | **p** | $-4 \times 4$ |

**2** Measure each angle. 2.02

**a**



**b**



**c**



**3** Draw an example of: 2.03

**a** an obtuse angle **b** an acute angle **c** a reflex angle

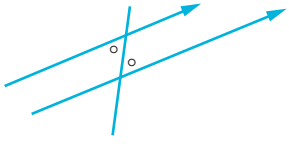
**4** Find the distance between each pair of numbers on a number line. 1.02

**a** $-1$ and $-3$ **b** $-5$ and $3$ **c** $0$ and $-4$

**5** Round each number to the nearest hundred. 3.01

**a** 2477 **b** 17 204 **c** 807 928

**6** Which type of angles are shown on the diagram? 2.10
Select the correct answer **A**, **B**, **C** or **D**.

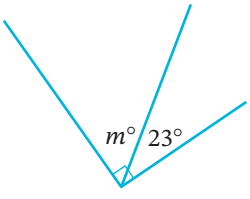


**A** alternate **B** co-interior
**C** corresponding **D** vertically opposite

**7** Evaluate each expression. 3.02

**a** $217 \times 7$ **b** $85 \times 22$ **c** $156 \div 6$

**8** Find the value of $m$. Select the correct answer **A**, **B**, **C** or **D**. 2.04




**A** 23 **B** 67
**C** 157 **D** 337

**9** Find the complement of: 2.04

**a** 66° **b** 85° **c** 12°

**10** Find the supplement of: 2.04

**a** 32° **b** 90° **c** 105°

**11** Use a divisibility test to test whether 1955 is divisible by: 3.04

**a** 3 **b** 4 **c** 5

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1.03 **12** Which statement is true? Select **A**, **B**, **C** or **D**.

**A** $-7 > 3$ **B** $8 < -11$ **C** $0 < -15$ **D** $5 > -9$

2.05 **13** Draw 2 intersecting lines and mark in a pair of vertically opposite angles.

2.10 **14** **a** Draw a pair of parallel lines and cut them with a transversal.

**b** Mark a pair of alternate angles with (•).

**c** Mark a pair of corresponding angles with (*).

**d** Mark a pair of co-interior angles with (+).

1.08 **15** Evaluate each expression.

**a** $-3 \times 2 - 5$ **b** $-2 - 3 \times 8$ **c** $12 - 14 \div 2 + 5$

**d** $5 - [2 + (-2 \times 6)]$ **e** $15 - 3 \times 5$ **f** $7 \times 3 + 2 \times 5$

**g** $26 \div 2 - 14 \div 7$ **h** $7 + 5 \times (12 - 3)$ **i** $[(3 + 5) \times 2 - (20 \div 5)] \times 5$

**j** $\dfrac{22 + 3}{-7 + 12}$ **k** $\dfrac{18 - 24}{7 - 4}$ **l** $\dfrac{15 - 18}{3} + 4$

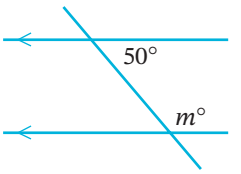
3.09 **16** Which number is NOT a factor of 66? Select **A**, **B**, **C** or **D**.

**A** 3 **B** 6 **C** 9 **D** 11

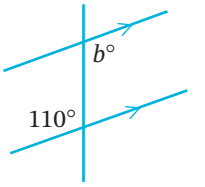
2.04 **17** Find the value of each variable.

2.05 2.10

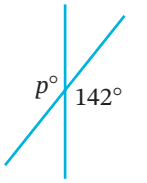
**a**



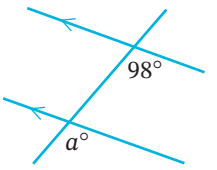

**b**



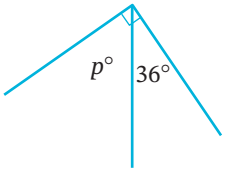

**c**



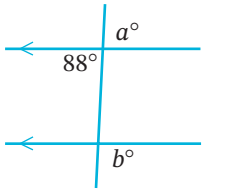

**d**



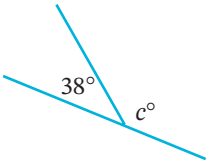

**e**



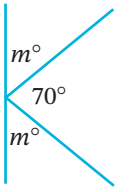

**f**



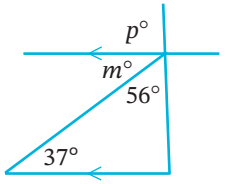

**g**



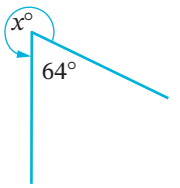

**h**



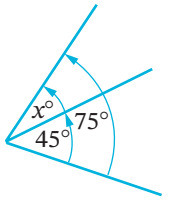

**i**



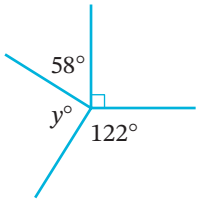

**j**




**k**




**l**




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**18** Evaluate each expression. 3.07-3.08

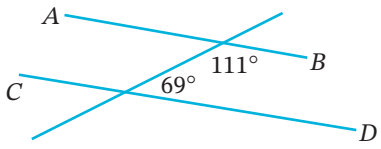
**a** $6^2$ **b** $3^3$ **c** $\sqrt{25}$ **d** $\sqrt{121}$

**19** The average daytime temperature in winter is 13°C and the average night-time temperature is −5°C. Find the difference between these 2 temperatures. 1.10

**20** Use a factor tree to write each number as a product of its prime factors. Write your answers using index notation. 3.10

**a** 18 **b** 45 **c** 360

**21** Is $AB \parallel CD$? Give a reason for your answer. 2.11




**22** Find the highest common factor of: 3.11

**a** 24 and 36 **b** 45 and 81